Feature Extraction by Common Spatial Pattern in Frequency Domain for Motor Imagery Tasks Classification

Jie Wang[1], Zuren Feng[1], Na lu[1]

1. State Key Laboratory for Manufacturing System Engineering, System Engineering Institute, Xi'an Jiaotong University, Xi'an 710049
E-mail: wangjiexjtu@163.com

Abstract: Common spatial pattern (CSP) as a feature extraction algorithm has been successfully applied to classify EEG based motor imagery tasks in brain computer interface (BCI). Successful application of CSP depends on the character of input signals and the first and last m eigenvectors of projection matrix. In this study, we proposed a novel and robust feature extraction method designated frequency domain CSP (FDCSP) that the samples in frequency domain obtained by fast Fourier transform (FFT) algorithm and evenly distributed in 8-30Hz were employed as the input signals of CSP. Besides, we made some modifications to classical CSP to address the inconsistent issue and enhance the generalization ability. Cross validation classification accuracy and standard deviation based on training data were employed as the principle to optimize the subject-specific parameter m. Two public EEG datasets (BCI competition Ⅳ dataset 2a and 2b) were used to validate the proposed method. Experimental results demonstrated that the proposed method significantly outperformed many other state-of-the-art methods in classification performance. What's more, samples in frequency domain as the input signals of CSP are demonstrated more robust against preprocessing. Based on the two public datasets, the proposed FDCSP method has potential significance to motor imagery based BCI design in practice.

Key Words: Common Spatial Pattern (CSP), Frequency Domain Samples, Motor Imagery Tasks Classification

1 INTRODUCTION

The EEG signals based brain computer interface (BCI) system is a communication channel for exchanging messages between the human brain and the computers [1]. It aims to translate brain signals into control commands for out devices [2], thereby building a direct connection between human brain and external environment. It has been proven that the imagination of movements will modulate brain electrical activities [3], and can be measured by EEG signals. Activation of motor area neurons either by a real movement or by imagination of the movement is similar to some extent [4]. Therefore, depending on the type of motor imagery, different measureable EEG patterns can be obtained. Based on different EEG signals, such as mu and beta rhythms, slow cortical potentials [5], event-related P300 [6, 7], visual evoked potentials [8] and so on, many kinds of BCIs systems were born. Among these BCI systems, the motor imagery based BCI systems stood out for its advanced approach for signal processing and low requirement for auxiliary equipment as well as its safety for noninvasive measurement. Therefore, the MI based BCIs systems have received widely research over the past decades.

The kernel parts of a BCI system are preprocessing, feature extraction and EEG pattern classification. To simply this problem, raw EEG signals can be considered as a linear mixture of source signals and noise [9]. Therefore, removing the noise is a very essential part before feature extraction. In the feature extraction part, common spatial pattern (CSP) is a popular spatial filtering algorithm and has been proven to be an effective feature extraction algorithm for the binary motor imagery tasks classification [10, 11]. However, it is hard to find the global optimal projection matrix because the difference in the time domain signals is week in comparison with in the frequency domain and the asynchronization exists in time domain. To overcome the drawbacks in time domain, several improved CSP algorithms were proposed, such as discriminative filter bank common spatial pattern (DFBCSP) [12] and filter bank common spatial pattern (FBCSP) [13]. The idea of SBCSP and FBCSP is to decompose the raw EEG signals into multiple frequency bands and employ the CSP on each band to select the most discriminative features. However, the decomposition of the EEG signals into narrow overlapping frequency bands will damage the raw signals seriously, and the classification accuracy will be affected in return. This drawback motivates us to decompose EEG signals into as fewer parts as possible in order to keep the high quality of EEG signals. That a wide filter band is adopted for CSP will minimize the damage to EEG. ERD and ERS show motor imagery could cause the energy of frequency rhythms increased or decreased in the related cortical areas, which illustrates the discriminative information is concentrated in the fixed frequency bands. Therefore, applying samples in frequency domain as the input signals will obtain better classification performance theoretically. Meanwhile, this kind of processing can overcome the asynchronization in time domain between input signals.

In this study, we propose a novel method termed FDCSP that imposes the samples in frequency domain obtained by FFT, especially distributed in the 8-30Hz frequency band.

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Concerned there were some inconsistent problems [14] in classical CSP, which will be discussed below, we made some modifications to solve this problem. In addition, before the feature extraction, we imposed the widely used 7-30Hz band-pass filter to purify the raw signals as the preprocessing, and the cross validation method was employed to optimize the subject dependent filters. After the feature extraction scheme, a series of training and testing feature vectors are obtained based on this proposed method. Finally, based on the training feature vectors, the SVM classifier with linear kernel is trained. To the best knowledge of authors, this is the first time that the modified CSP in this way and input samples in frequency domain by FFT are extended and introduced into BCI, which enables the pattern analysis of brain signals in a new appropriate way. The proposed FDCSP method is validated on two public EEG dataset (BCI competition Ⅳ dataset 2a and 2b), and the experimental results indicate this new method is more robust and achieves ensemble higher classification accuracy in comparison with classical CSP and other advanced CSP methods.

The main contributions of this work are as follows: Firstly, a detailed study on MI tasks classification in this paper demonstrates the samples in frequency domain as the input achieve significant better classification performance. Recently proposed DFBCSP [12] and FBCSP are based on sub-bands decomposition with multiple filters that have different pass-bands. Although these methods can also achieve higher classification accuracy, the selection of feature values is time consuming and do not give enough consideration to frequency samples. Secondly, samples in frequency domain as the input are more robust against preprocessing. Without the procedure of preprocessing, it will have an obvious negative impact on the classification accuracy, which will be deeply discussed below. However, the frequency samples as the input are less affected. It should be also noted that this calculation model will be more significant to online BCI application. Thirdly, the modification we made for classical CSP method is mainly reflected in the estimation of feature matrix and feature vectors. It should be noted that while there have been several approaches of variations of CSP through more robust covariance matrix estimation [15, 16]. Regularized CSP (R-CSP) and other robust CSP variations are willing to solve small training sample problems and lower the effect of noise and overcome overfitting, which will be compared in this paper.

The rest of this paper is structured as follows: In section 2, we made a brief introduction to classical CSP and its application in classifying MI tasks. Further, we detailed the modification to CSP and how to use frequency samples by FFT as the input of CSP. In section 3, we presented how to process the open datasets we used in this study. Section 4 presents the experimental results. The conclusion was shown in section 5.

2 METHOD

In this section, on one hand, basic concepts of the classical CSP and its application to EEG feature extraction were described and we detailed its variant that we proposed. On the other hand, we would explain how to use the samples in frequency domain as the input of CSP to extract feature vectors.

EEG records brain activities as multichannel time series from multiple electrodes placed on the scalp of a subject. Let $\mathbf{X} \in R^{N \times T}$ presented the filtered EEG signal, where N is the number of channels and T is the number of samples.

2.1 Feature Extraction by Classical CSP

The sample covariance matrix of a trial is calculated as follows:

$$\mathbf{R} = \frac{\mathbf{X}\mathbf{X}^{\mathbf{T}}}{tr(\mathbf{X}\mathbf{X}^{\mathbf{T}})} \tag{1}$$

where the superscript "T" denotes the transpose of a matrix, and $tr(\mathbf{x})$ denotes the trace of a matrix. Then, calculate the composite spatial covariance as follows:

$$\overline{\mathbf{R}}_1 + \overline{\mathbf{R}}_2 = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathbf{T}} \tag{2}$$

where $\mathbf{U}$ denotes the matrix of eigenvectors, and $\mathbf{\Lambda}$ denotes the diagonal matrix of corresponding eigenvalues. The full projection matrix is then formed as follow:

$$\mathbf{W} = \mathbf{B}^{\mathbf{T}}\mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{U}^{\mathbf{T}} \tag{3}$$

where $\mathbf{B}$ denotes the matrix of eigenvectors for the whitened spatial covariance matrix. Only the first and last m eigenvectors corresponding to the first and last m eigenvalues sorted in descending order were employed.

$$\mathbf{Z} = \mathbf{W}^{\mathbf{T}}\mathbf{X} \tag{4}$$

A 2m dimensional feature vector is then constructed from the variance of the rows of $\mathbf{Z}$ as follows:

$$\mathbf{f_q} = \log\left(\frac{\mathrm{var}(\mathbf{z_q})}{\sum_{i=1}^{2m}\mathrm{var}(\mathbf{z_i})}\right) \tag{5}$$

where $\mathbf{z_q}$ denotes the q-th row vector of $\mathbf{Z}$.

2.2 The Modification to Classical CSP

There are several approaches to proposing variations of CSP through more robust covariance matrix estimation and feature extraction. Among these variations, regularized common spatial pattern (R-CSP) algorithms stand out for their robustness to artifacts and classification accuracy improvement with small training samples and overcoming overfitting. The modification we made for classical CSP is also inspired by the principle of R-CSPs. From the aforementioned, the estimation of covariance matrix (1) is not consistent with (4). To address these problems, we make two modifications to CSP to further regularized CSP in projection procedure, and on the other hand we make the covariance matrix (1) consistent with (4). The estimation of covariance matrix (1) guarantees that the sum of the energies of all signal channels is equal to 1 and thus makes it more robust against possible variations in the amplification degree of the signal. Therefore, we need change the (4) as follows:

$$\mathbf{Z} = \mathbf{W}^{\mathrm{T}}\mathbf{X} / \sqrt{tr(\mathbf{X}\mathbf{X}^{\mathrm{T}})} \tag{6}$$

In the end, the step (5) was modified. This step makes feature vectors more intuitive to build hyperplane. The modification was further described as follows:

$$\mathbf{f} = [\log(\mathrm{var}(\mathbf{z}_1)), \log(\mathrm{var}(\mathbf{z}_2)), \cdots, \log(\mathbf{z}_{2m})]^T \tag{7}$$

The feature vectors obtained above can be used to train the SVM classifier with simple linear kernel for motor imagery classification. The selection of parameter m was determined by the cross validation on training data. The Figure 1 and Figure 2 depict the distributions of the most significant two features derived by classical CSP and modified CSP in our way respectively from B0803T of dataset 2b. Our experimental results show the classification accuracy improvement only by this modification is 2.5% higher than classical CSP. We should also note that it is easier to recognize the location of every trial, which is of great importance.

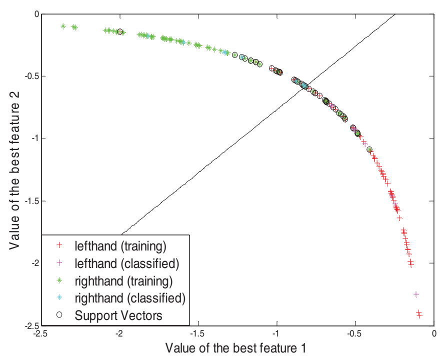

Fig 1. Distribution of the most significant two features by classical CSP from B0803T by SVM. Every point stands for a trial mapped into a 2 dimensional space (m=1).

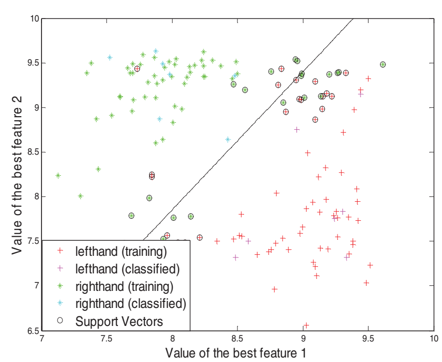

Fig 2. Distribution of the most significant two features by modified CSP from B0803T by SVM. Every point stands for a trial mapped into a 2 dimensional space (m=1).

2.3 Frequency Domain Transformation

Sensor stimulation, motor behavior, and mental imagery can change the functional connectivity within the cortex and result in an amplitude suppression or enhancement of mu and central beta rhythms [17, 18]. The discriminative information is mainly reflected in the energy increased or decreased in some specific frequency sub-bands, which proves the difference between two kinds of motor imagery is higher in the frequency domain than that in the time domain. Therefore, compared with common samples in time domain, the samples in mu and central beta frequency domain are more distinguishable and synchronized. As it has been shown that this three electrodes, C3 C4 Cz, which located in the left, central and right motor cortex of the brain respectively, are sufficient to obtain MI classification accuracy [10]. To quantify the discriminative information between left and right hand motor imageries, the correlation coefficients are employed, and in this study we calculate and figure the correlation coefficients during MI period for all subjects (from dataset 2a) from channel Cz. Figure 4 shows the correlation coefficients between left and right hand motor imageries for nine subjects in time and frequency domain respectively. It can be clearly illustrated that the correlation coefficients in time domain are relatively higher than those in frequency domain except for the subject8.

After the data preprocessing procedure, we can assume most of noise and artifacts are removed without too much damage to EEG signals. Then, FFT algorithm is used for domain transform and we transform time domain signals into frequency domain signals.

$$\widehat{\mathbf{X}} = \mathrm{FFT}(\mathbf{X}) \tag{8}$$

Theoretically, the values of each row of $\widehat{\mathbf{X}}$ are evenly distributed within 0~fs symmetrically. In this study, we cut off the specific sub-band within 8-30Hz signals. We assume the number of samples distributed within 8~30Hz is M , and the dimension of input signals in frequency domain is $\mathrm{N} \times \mathrm{M}$. In the end, the frequency samples obtained by this way can be used as the input in the feature extraction.

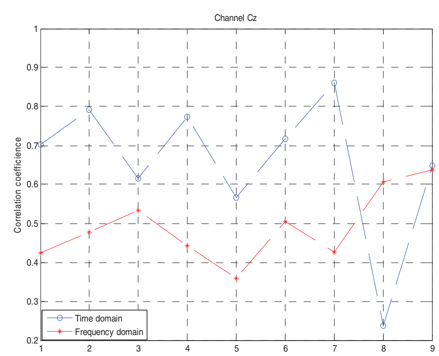

Fig 3. The correlation coefficients of channel Cz between left hand and right hand motor imagery in time domain and frequency domain. The blue symbols o stand for the correlation coefficients of nine subjects in time domain, and the res symbols * are in frequency domain.

3 Experimental Study

The dataset 2a comprises EEG signals from 9 subjects using 22 channels, which were recorded and sampled with sampling rate of 250Hz and band-pass filtered between

0.5Hz and 100Hz. The cue-based BCI paradigm consists of four different motor imagery tasks, namely the imagination of movement of the left hand, right hand, both feet and tongue. However, only two class MI tasks of left and right hand are considered. There are two sessions held, and each session has 72 trials for each MI task. On each trial, a time segment signal is extracted from 0.5 to 2.5s after the visual cue, as recommended by the winner of BCI competition IV. A wide filter band 7-30Hz is adopted before the feature extraction.

The dataset 2b is recorded from 9 subjects at electrodes C3, Cz and C4 with sampling rate of 250Hz. For comparison with the results reported in other literatures [19], this study only uses the third training session of the dataset, i.e., "B0103T", "B0203T",…,"B0903T". For each subject, a total 160 trials of EEG measurements are available (half for each class of MI). On each trial, a segment is extracted starting from 0.5 to 2.5s after the visual cue. A wide filter band 7-30Hz is adopted before the feature extraction.

The effectiveness of the proposed FDCSP method is tested on the two above mentioned datasets. To obtain the parameter m in dataset 2a, 8-fold cross validation was applied to select robust m. Because there are 72 trials in total for training and the same for evaluating. In common spatial pattern algorithm, m is a critical uncertain parameter ranged from 1 to 11 and decides the dimension of feature vectors. Note that the parameter searching procedure must be based on training data. Cross validation accuracy (CVA) is the percentage of data which are correctly classified based on training data. In this paper, we apply a "grid search" [20] on m using cross validation in which various m values are tried and the one with the best cross validation accuracy is selected. After obtaining the selected parameter m, we use SVM classifier with linear kernel function to classify testing trials. For performance evaluation, the correct classification rate (CCR) is used to measure the classification accuracy for dataset 2a in the end. It also should be noted that there are 3 channels in total in dataset 2b. Parameter selection is not appropriate. During the evaluation of proposed method, the parameter m is set at 1. The competition winner used the filter bank CSP technique for feature extraction along with the Naive Bays Parzen window classifier. Therefore, it is essential to compare our proposed method with FBCSP and other advanced methods. For dataset 2b, there are 80 trials for each MI task in one session, so a 10-fold CVA is implemented to evaluate the classification performance as in [19].

4 EXPERIMENTAL RESULTS

For the performance evaluation, the methods explained in the previous section are applied accordingly. The proposed method has been evaluated using the dataset 2a and 2b. The results obtained by the proposed method are compared with several state-of-the-art methods which are presented in Table 2 and Table 3 for dataset 2a and 2b respectively.

Based on the results presented in Table 1, only the first 3 subjects were chosen for an example, the optimized parameter m is subject-specific. For every subject, we select the parameter m corresponding to the highest cross validation accuracy, which was highlighted with bold fonts. The results obtained in Table 1 were carried out for 1000 times 8-fold cross validation, and the standard derivation was also an important index to select parameter m. For example, m=8 and m=9 achieved the highest accuracy for subject 2, but m=8 was selected because of its smaller standard derivation. According to this principle, the best subject-specific parameter m was decided for every subject.

Table1. Cross validation accuracy in m selection for dataset 2a from first 3 subjects.

m	A01	A02	A03
1	0.8850(0.0054)	0.7403(0.0123)	0.9819(0.0036)
2	0.9572(0.0061)	0.8160(0.0110)	0.9792(0.0065)
3	**0.9858(0.0049)**	0.8549(0.0061)	0.9757(0.0059)
4	0.9771(0.0106)	0.8979(0.0104)	0.9792(0.0087)
5	0.9661(0.0087)	0.8840(0.0167)	0.9868(0.0039)
6	0.9716(0.0090)	0.9076(0.0098)	**0.9938(0.0022)**
7	0.9691(0.0103)	0.9250(0.0091)	0.9875(0.0072)
8	0.9691(0.0095)	**0.9299(0.0155)**	0.9687(0.0059)
9	0.9717(0.0099)	0.9299(0.0172)	0.9701(0.0074)
10	0.9670(0.0088)	0.9187(0.0114)	0.9757(0.0059)
11	0.9776(0.0083)	0.9139(0.0123)	0.9743(0.0074)

In table 2, the results of proposed FDCSP method evaluated on dataset 2a are compared with classical CSP, Sparse CSP pattern (SCSP) [21], and regularized CSP methods [16]. In regularized CSP methods, a Tikhonov regularized CSP (TRCSP) and a weighted regularized CSP (WTRCSP) were selected from all R-CSP methods. TRCSP is the best feature extraction algorithm, as it reached both the best CCR and standard deviation. The highest CCR for every subject was in bold front. It can be easily found that all of the four advanced CSP algorithms outperformed the classical CSP. The proposed FDCSP method further improve the ensemble CCR than classical CSP, with an average improvement of 4.9%. The results in Table 2 also clearly suggests that there is a significant improvement in the classification accuracy for dataset 2a of at least 1.3% over the existing SCSP method which itself is proven to be better than RCSP and SVM [21].

Table 3 summarizes the CVA for the nine subjects from dataset 2b. For every subject, the best CVA was highlighted with bold front. The proposed FDCSP method yielded lower classification errors for most subjects than those of the other three methods. It clearly indicates that all of the three methods (FDCSP, FBCSP and DFBCSP) outperformed the classical CSP in classification performance at least 2.4%. The proposed FDCSP method further improve ensemble CVA than classical CSP, with an average improvement of 4.39%. The average CVA improvement achieved by the FDCSP method were 1.9% and 0.6% in comparison with the FBCSP and DFBCSP methods respectively. Experimental results without preprocessing procedure are also calculated. Compared with the experiments of Table 2 and Table 3, the sole difference is that we conduct experiments without eliminating noise and artifacts by a band-pass filter. However, the noise and artifacts will damage essential components of input signals so that the classification accuracy will be severely affected. Classical CSP algorithm processes the input signals in time domain directly and will obtain awful classification results with an average of 8.57% and 5.13% less than common calculation model. However, our proposed method was less affected by preprocessing. CSP algorithm in frequency domain makes full use of the frequency signals distributed between 8-30Hz so that the frequency samples are less affected, which makes FDCSP relatively more robust against preprocessing. Compared with common calculation model, FDCSP without preprocessing is almost similar with 1.62% less. When preprocessing procedure is eliminated, FDCSP method significantly improves the classification accuracy in comparison with classical CSP ($p \leq 0.01$). Therefore, this study can also indicate that FDCSP is more robust and can extract distinguishable feature vectors from raw EEG signals to reduce calculation consumption in high order band-pass filters, which is more beneficial to online BCI application.

Table2. The CCR for dataset 2a in comparison with the state-of-the-art methods.

Subject	FDCSP	CSP	SCSP	TRCSP	WTRCSP
A01	**0.9166**	0.8889	**0.9166**	0.8889	0.8889
A02	**0.6805**	0.5139	0.6736	0.5417	0.5486
A03	0.9722	0.9653	**0.9791**	0.9653	0.9653
A04	**0.7222**	0.7014	**0.7222**	0.7083	0.7014
A05	**0.7222**	0.5486	0.6527	0.6250	0.6597
A06	**0.7153**	**0.7153**	0.6627	0.6736	0.6181
A07	0.8125	0.8125	**0.8472**	0.8125	0.8125
A08	**0.9861**	0.9375	0.9722	0.9587	0.9583
A09	**0.9375**	**0.9375**	0.9166	0.9167	0.9097
Average	**0.8294**	0.7801	0.8163	0.7879	0.7847

Table3. The CVA for dataset 2b in comparison with the state-of-the-art methods.

Subject	FDCSP	CSP	FBCSP	DFBCSP
B0103T	**0.7813**	0.7656	0.7750	0.7794
B0203T	**0.6903**	0.5556	0.5594	0.5725
B0303T	**0.5819**	0.5262	0.5375	0.5519
B0403T	**0.9938**	0.9806	0.9887	0.9881
B0503T	0.8688	0.8819	0.9044	**0.9244**
B0603T	0.7937	0.6937	0.7806	**0.8169**
B0703T	**0.9063**	0.8344	0.8650	0.8950
B0803T	0.8812	0.8656	**0.8875**	0.8862
B0903T	0.8187	0.81750	0.8388	**0.8475**
Average	**0.8129**	0.7690	0.7930	0.8069

5 CONCLUSION

In this study, we introduced a modified CSP in frequency domain termed FDCSP algorithm to improve motor imagery tasks classification performance in BCI systems. We combined this modified CSP and samples in frequency domain together to achieve the highest classification accuracy in comparison with many state-of-the-art methods. In the proposed method, samples in frequency domain distributed within 8-30Hz by FFT algorithm were first employed as the input signals of CSP. Significant feature vectors were then extracted by this proposed method and were used for training SVM to classify MI tasks. Unknown parameter m in CSP were determined by cross validation accuracy and standard deviation based on training data. Experimental results based on two public EEG datasets (BCI competition Ⅳ dataset 2a and 2b) indicated our proposed FDCSP method yielded remarkable higher overall classification accuracy. It should be also stressed that the CSP feature extraction calculated by frequency samples were more robust against preprocessing procedure, which is more significant to real BCI systems.

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